HIGH LEVEL DESIGN

Soil sensor

Revision B · 2026-11-05

Made by trace from the project's records at commit `a41e9d0c2b17`

Every figure and value in this document names the record it came from. A value marked pending has not been worked out yet.

Revisions

Revision	Date	Commit	What changed
A	2026-10-21	`9c02e1f`	First release
B	2026-11-05	`a41e9d0`	Released

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1 Overview

The soil sensor measures soil moisture and temperature in a field and reports them over a long-range radio (LoRa) link to a gateway every 15 minutes. It runs for two years on two AA cells and sits in a sealed printed case on a stake.

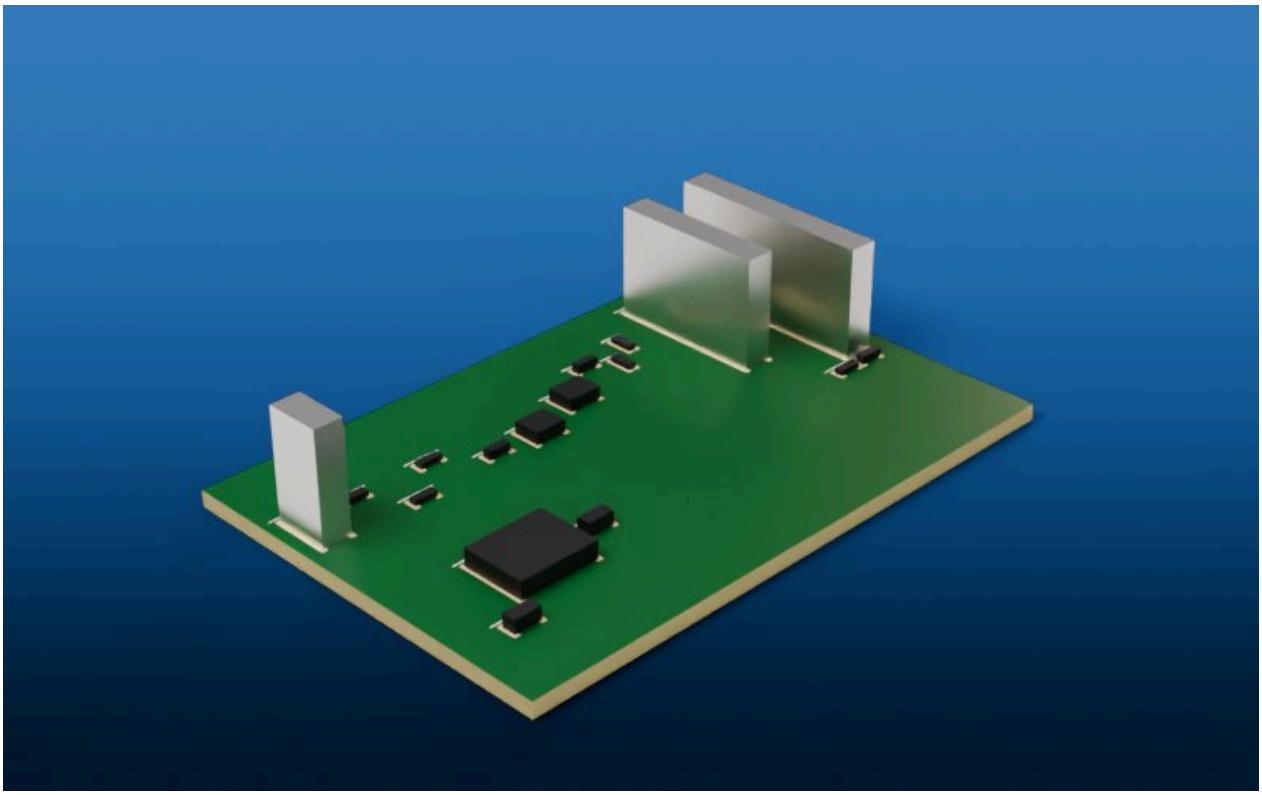


Figure 1: The populated board, Rev B, from the front left.

2 System context

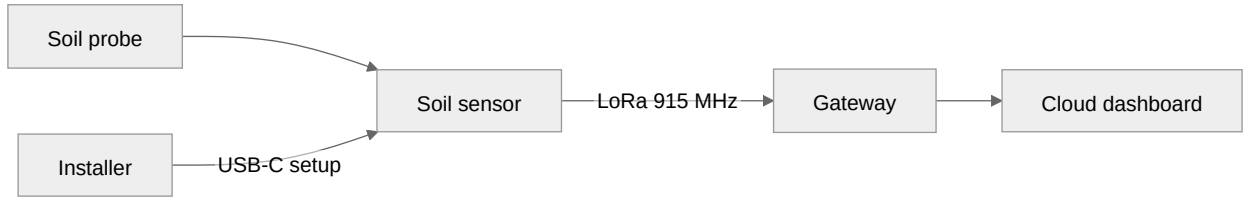


Figure 2: What the sensor talks to: the probe in the soil, the gateway over the radio, and a laptop over USB-C when it is set up.

3 Requirements

Id	Requirement	Status
R-001	Reports moisture within 3 % volumetric water content	accepted
R-002	Runs two years on two AA cells	accepted
R-004	Runs from one lithium cell, 3.3 V out	verified
R-006	Reaches a gateway 2 km away in open field	accepted

4 Architecture

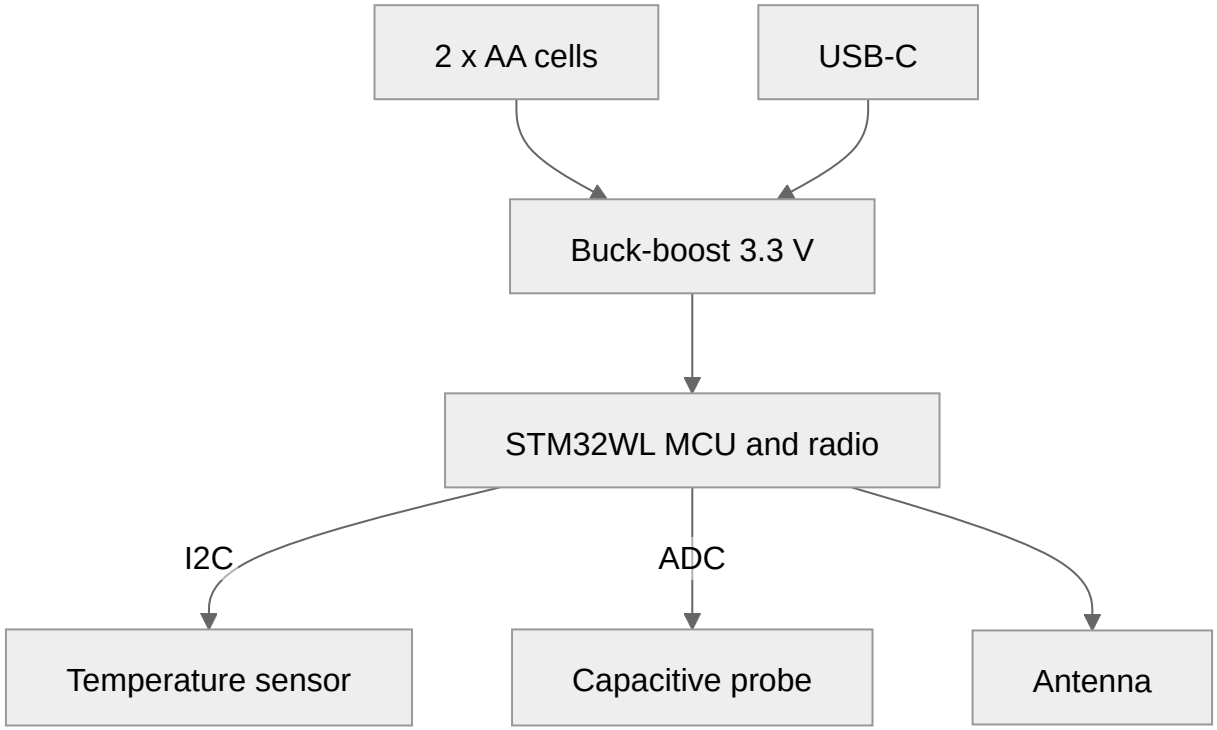


Figure 3: The board's blocks: power from the cells or USB-C, one microcontroller with its own radio, and the two sensors.

5 Data flow

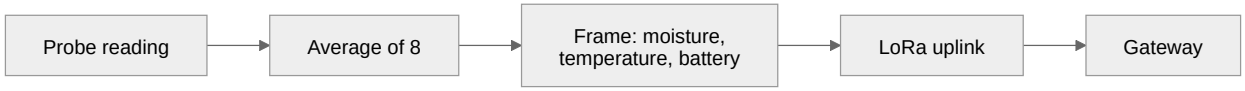


Figure 4: How one reading becomes one uplink every 15 minutes.

6 Key parts

Pending: The key parts, why each was chosen, and their stock.

7 Power and thermal summary

Pending: The power budget at a glance and the tightest margins.

8 Mechanical summary

Pending: Size, mounting and the case, with the in-case render.

9 Journey

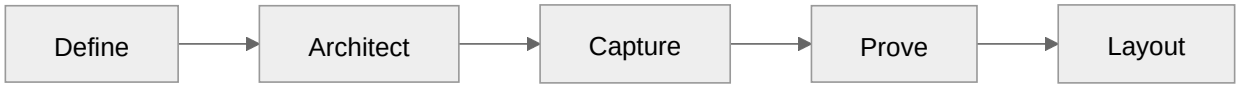


Figure 5: Where the project is: Capture, with the schematic being finished.

10 Decisions

- **One chip for the controller and the radio** (`STM32WL55`): fewer parts and lower sleep current than a separate radio.
- **Buck-boost regulator** (`TPS63900`): keeps 3.3 V from fresh cells down to 1.8 V.

11 Risks and open questions

Pending: What could still go wrong and who owns each open question.

12 Glossary

Pending: Every acronym and term used in this document.